On the other hand, if the braneworld scenario is correct, upcoming accelerator experiments *do* have the potential of confirming string theory. As mentioned briefly in Chapter 13, should the extra dimensions in the braneworld scenario be as large as 10^{-16} centimeters, not only would gravity be intrinsically stronger than previously thought, but strings would be significantly longer as well. Since longer strings are less stiff, they require less energy to vibrate. Whereas in the conventional string framework, string vibrational patterns would have energies that are more than a million billion times beyond our experimental reach, in the braneworld scenario the energies of string vibrational patterns could be as low as a *thousand* times the proton's mass. Should this be the case, high-energy collisions at the Large Hadron Collider will be akin to a well-hit golf ball ricocheting around the inside of a piano; the collisions will have enough energy to excite many "octaves" of string vibrational patterns. Experimenters would detect a panoply of new, never before seen particles— new, never before seen string vibrational patterns, that is —whose energies would correspond to the harmonic resonances of string theory.

The properties of these particles and the relationships between them would show unmistakably that they're all part of the same cosmic score, that they're all different but related notes, that they're all distinct vibrational patterns of a single kind of object—a string. For the foreseeable future, this is the most likely scenario for a direct confirmation of string theory.

Cosmic Origins

As we saw in earlier chapters, the cosmic microwave background radiation has played a dominant role in cosmological research since its discovery in the mid-1960s. The reason is clear: in the early stages of the universe, space was filled with a bath of electrically charged particles—electrons and protons—which, through the electromagnetic force, incessantly buffeted photons this way and that. But by a mere 300,000 years after the bang (ATB), the universe cooled off just enough for electrons and protons to combine into electrically neutral atoms—and from this moment onward, the radiation has traveled throughout space,